

Mastering Image Generation with Stable Diffusion

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DEVOXX™ POLAND

Platinum Sponsors: GRAND PARADE Red Hat

#DevoxxPL

Yesterday in Krakow



Image not generated by AI

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Senior Architect, Open Source Expert

WORLDLINE



- Worldline since 1999
- Based in Paris
- **Yoga Teacher**

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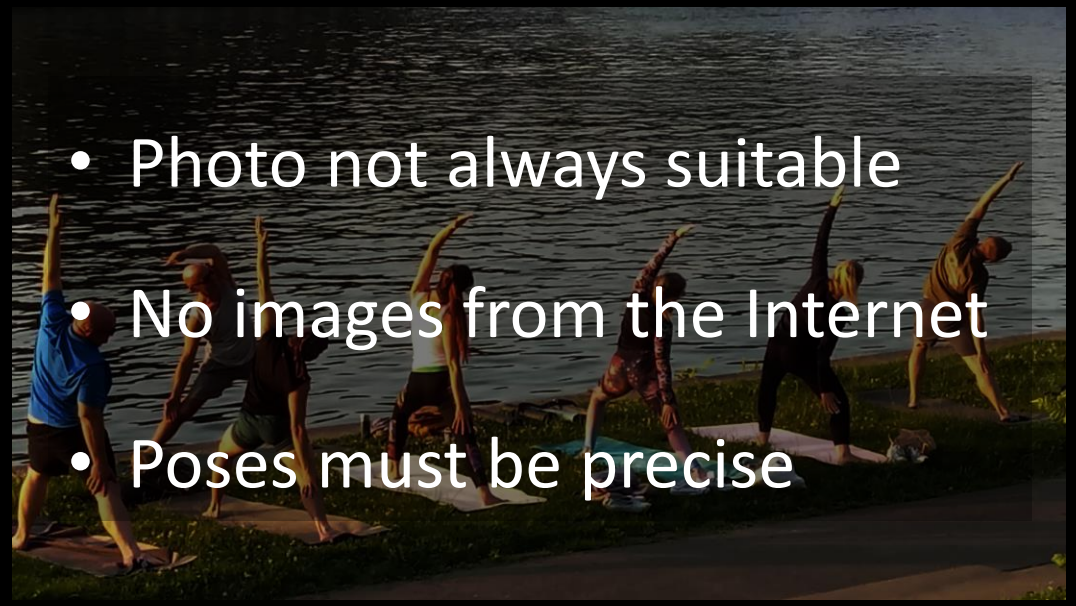
raphiki.bsky.social

raphiki.github.io

Use Case

Generate images of Yoga poses and sequences

YogĀrkana



- Photo not always suitable
- No images from the Internet
- Poses must be precise

How could Generative AI help?



From German labs to the London-based startup

- **Collaboration** between labs and companies
  - Latent Diffusion model in 2021
  - Prompt-guided streaming
  - Using the LIAON dataset
  - Participation of Runway and EleutherAI
- **Stability AI**
  - Donation of computing power to the project
  - Hiring most of the initial investigators
  - Now maintainer of Stable Diffusion models



runway



stability.ai

"Open" licenses

- Responsible AI: OpenRAIL
- Version 3.5: 1M+ revenue companies must pay



Vibrant Contributing Communities

Models

- Fine-tuning: models adapted to specific styles or themes
- Refiners, Upscalers, ControlNets
- Model Patches (LoRA)

Tools

- GUI: Automatic1111 Web UI, ComfyUI
- Fine-tuning Tools: Dreambooth, Kohya SS

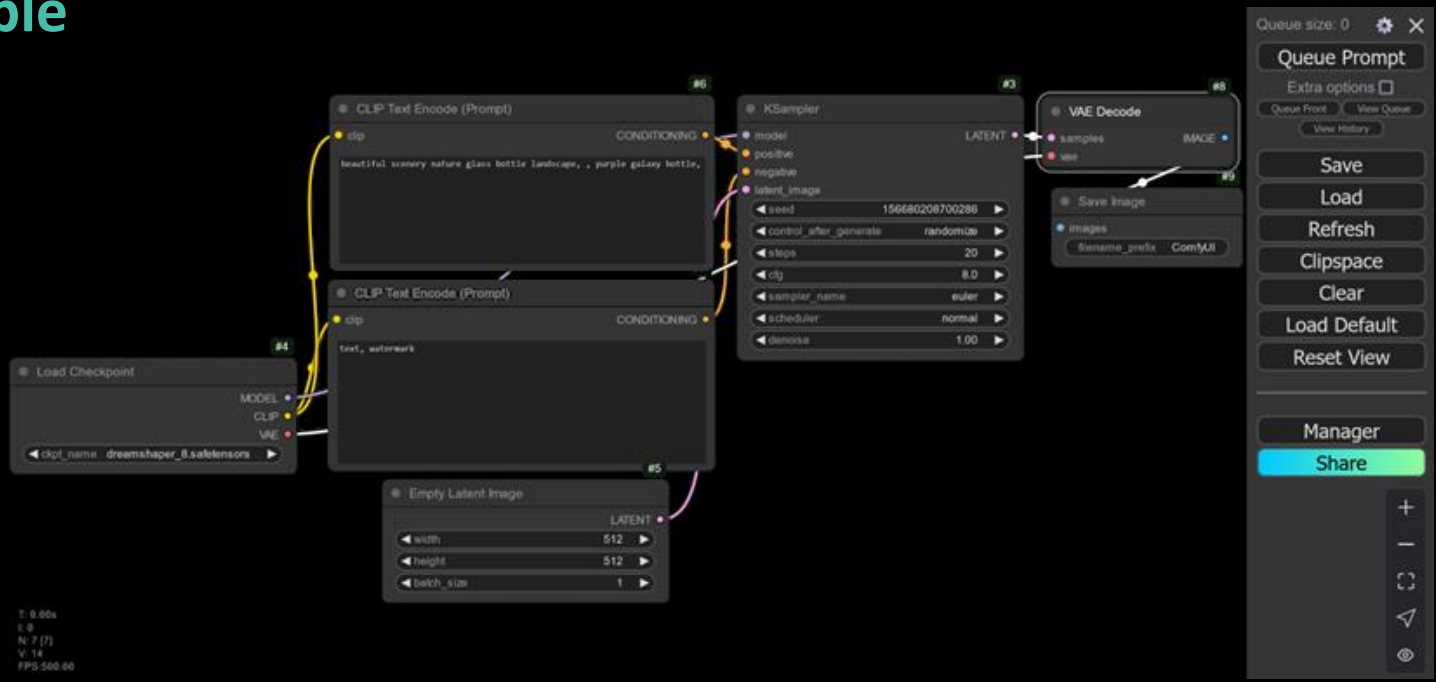
Creative Communities

- Sharing models, prompts, images and tutorials: Hugging Face, Civitai.com...
- Stable Horde: distributed cluster of build nodes

Comfy

Local GUI for Stable Diffusion workflows

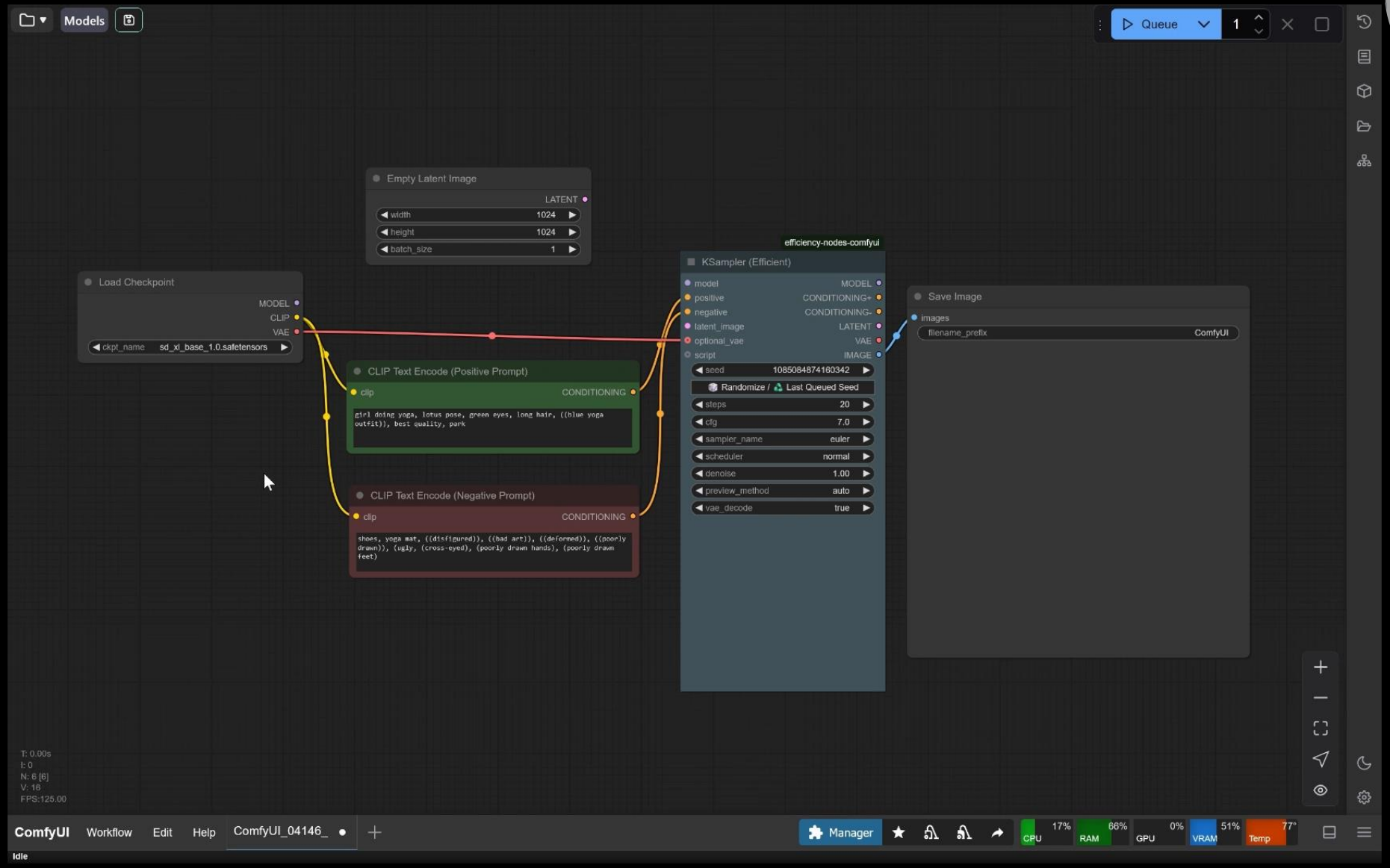
- **Intuitive, modular and customizable**
- **Flexible node-based workflows**
  - Text-to-Image Generation
  - Image-to-Image Processing
  - Custom nodes
- **Community aspect**
  - GPLv3 license
  - Contributions, plugins, doc



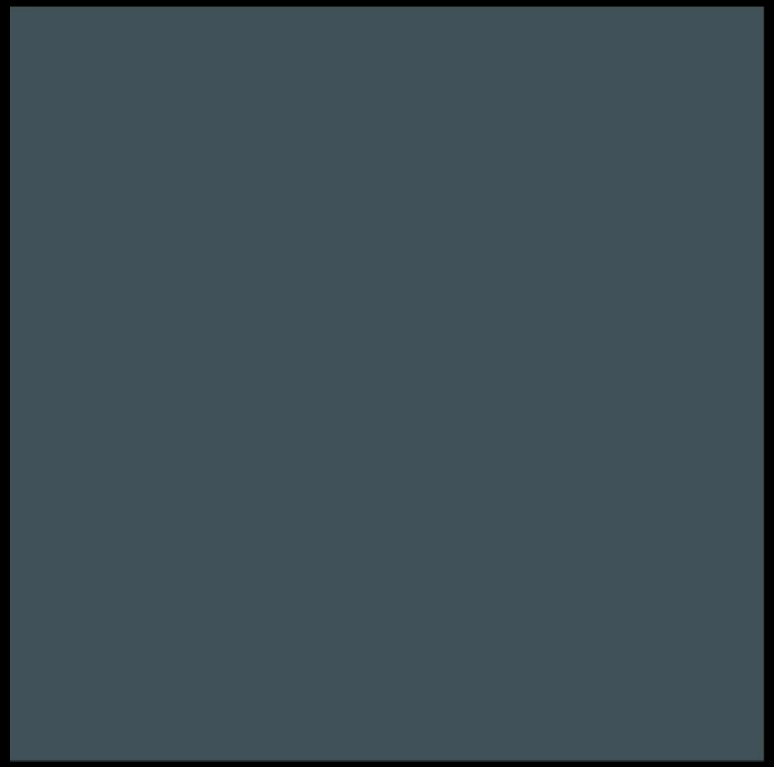
Let's start with a simple demo

Generate an image of a
woman doing yoga

Demo – Text-to-Image (T2I)



How does Stable Diffusion work?

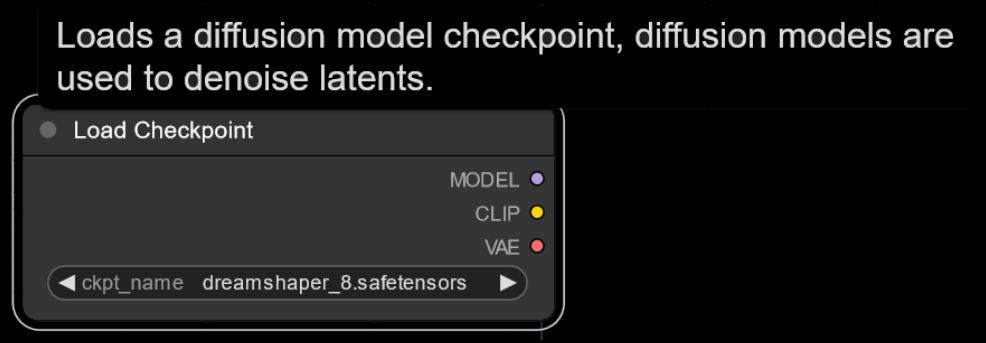


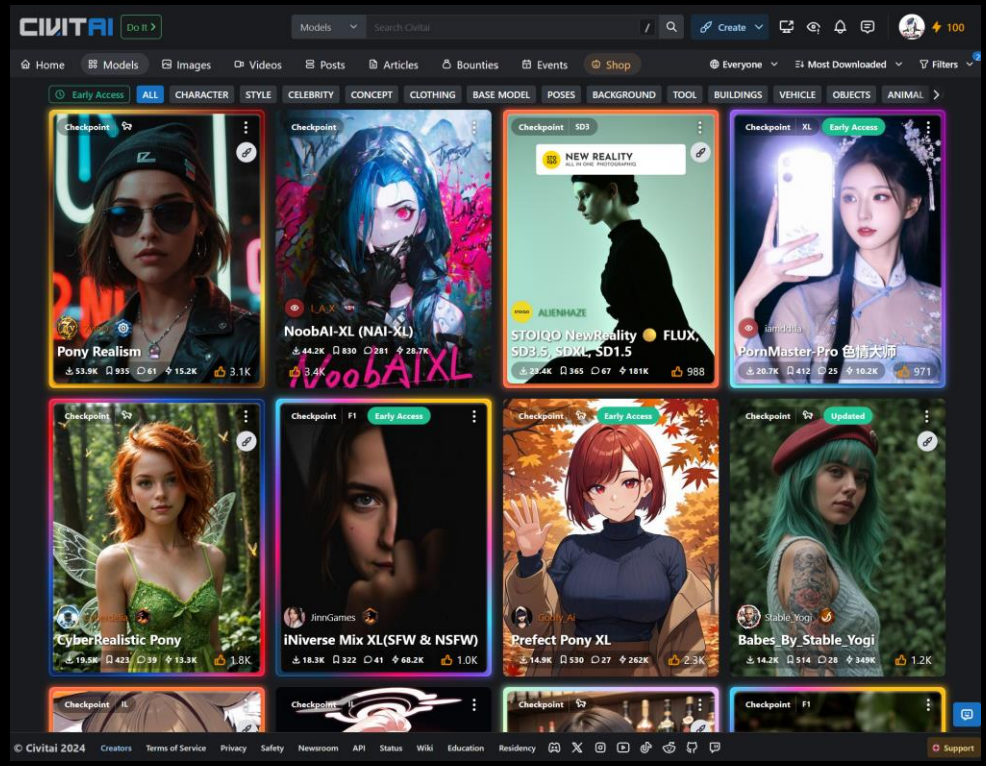
- Starts with a random noise: unrecognizable image
- Refines step by step: gradually suppresses noise by adding details
- Learns from real images: injects patterns from trained images
- Is text-driven: prompts like "Woman doing yoga in a park“
- Produces a final image: detailed and aligned with the prompt

Models

- **Most used Stability.ai models**
  - SD 1.5
  - SDXL
- **Fine-tuned Models**
  - Specialized: style, subject
  - Community (cf. civitai.com)

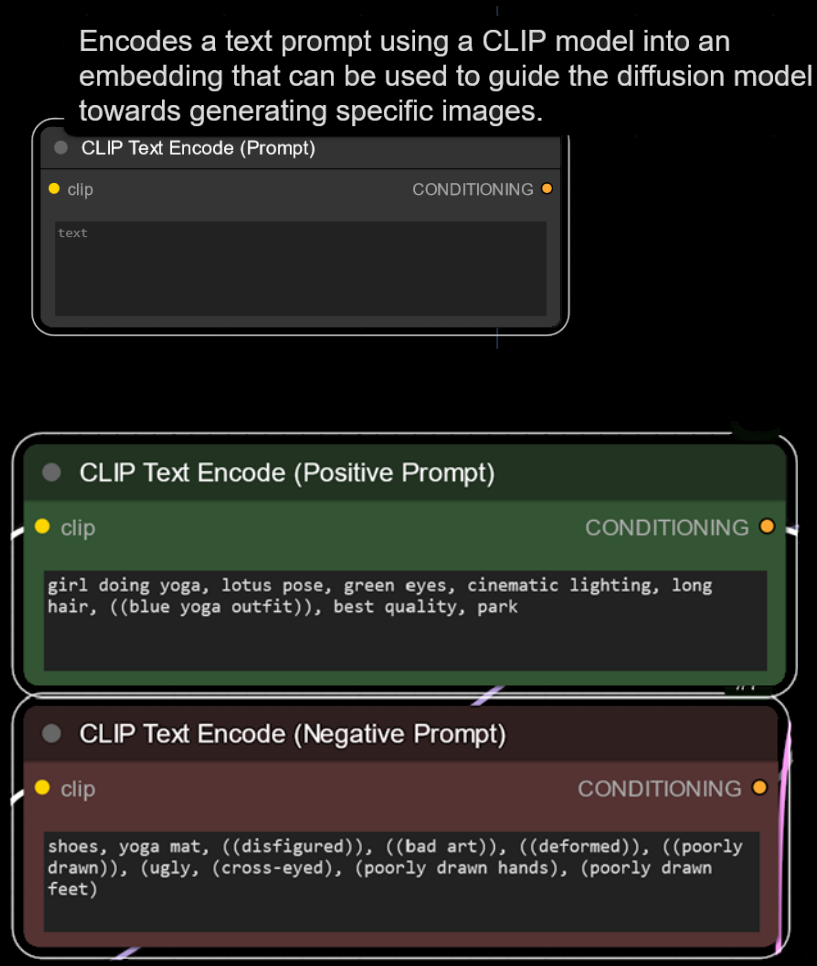
CHARACTER STYLE CELEBRITY CONCEPT CLOTHING BASE MODEL POSES
BACKGROUND TOOL BUILDINGS VEHICLE OBJECTS ANIMAL ASSETS ACTION





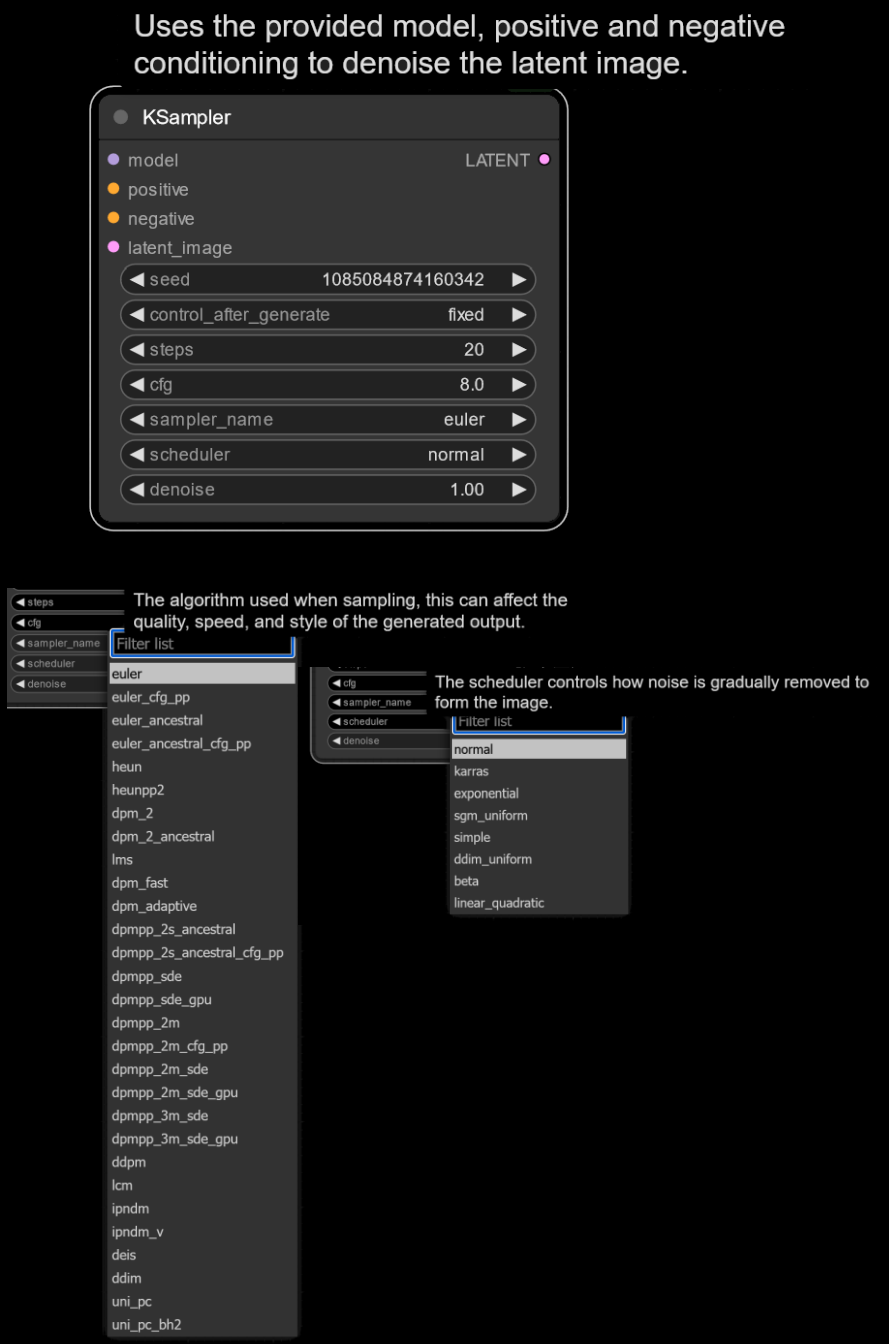
Prompts

- **CLIP Model** (Contrastive Language–Image Pretraining)
  - Combines descriptive text and images
  - Helps generate images for specific prompts
  - Can handle a wide range of prompts
  - Developed by OpenAI in 2021
  - OpenCLIP version by LAION
- **Positive and negative**
- **Textual or short syntax**



Denoising process

- **Seed**
  - Random element used to create the initial noise
  - Fixing it allows to check impact of other parameters
- **Samplers**
  - Algorithms guiding Iterative Image Generation
  - Differ in speed and quality
- **Schedulers**
  - Control how noise is eliminated at each stage
  - Also impact speed and quality
- **Other settings**
  - # steps, CFG (prompt impact), % denoising



We can tweak the prompt



long hair,
((blue yoga outfit))



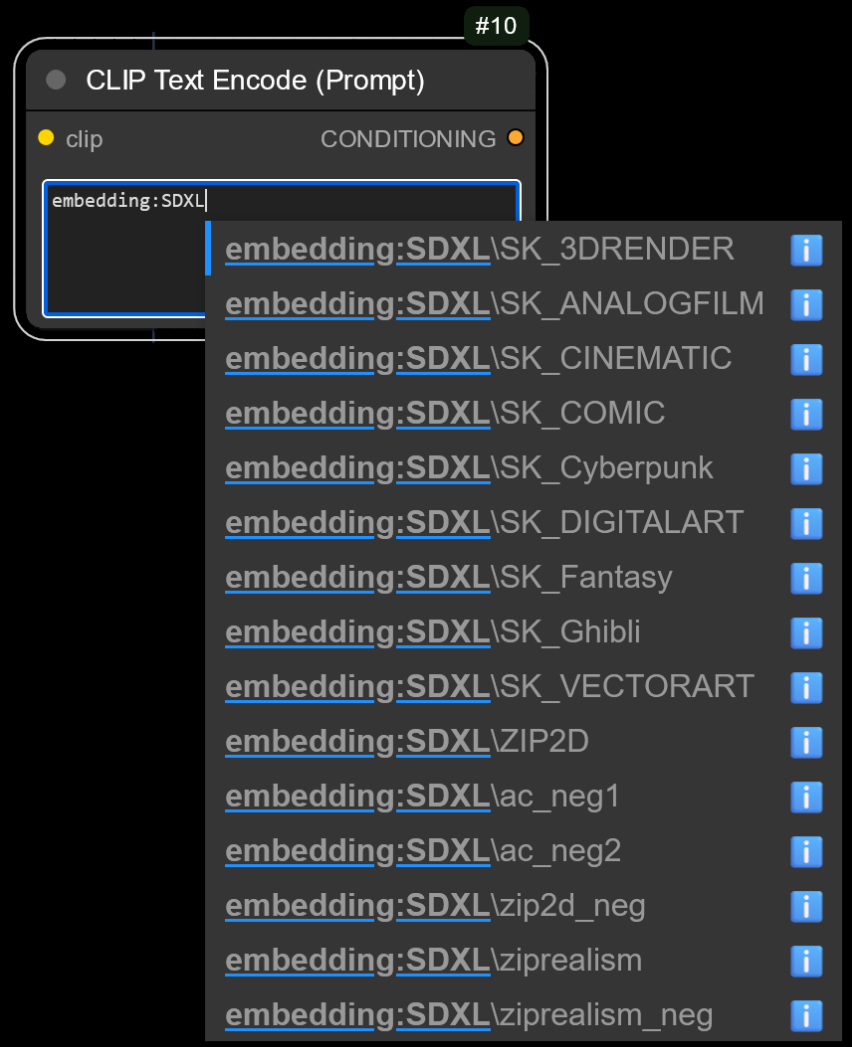
long red hair,
((pink yoga outfit))



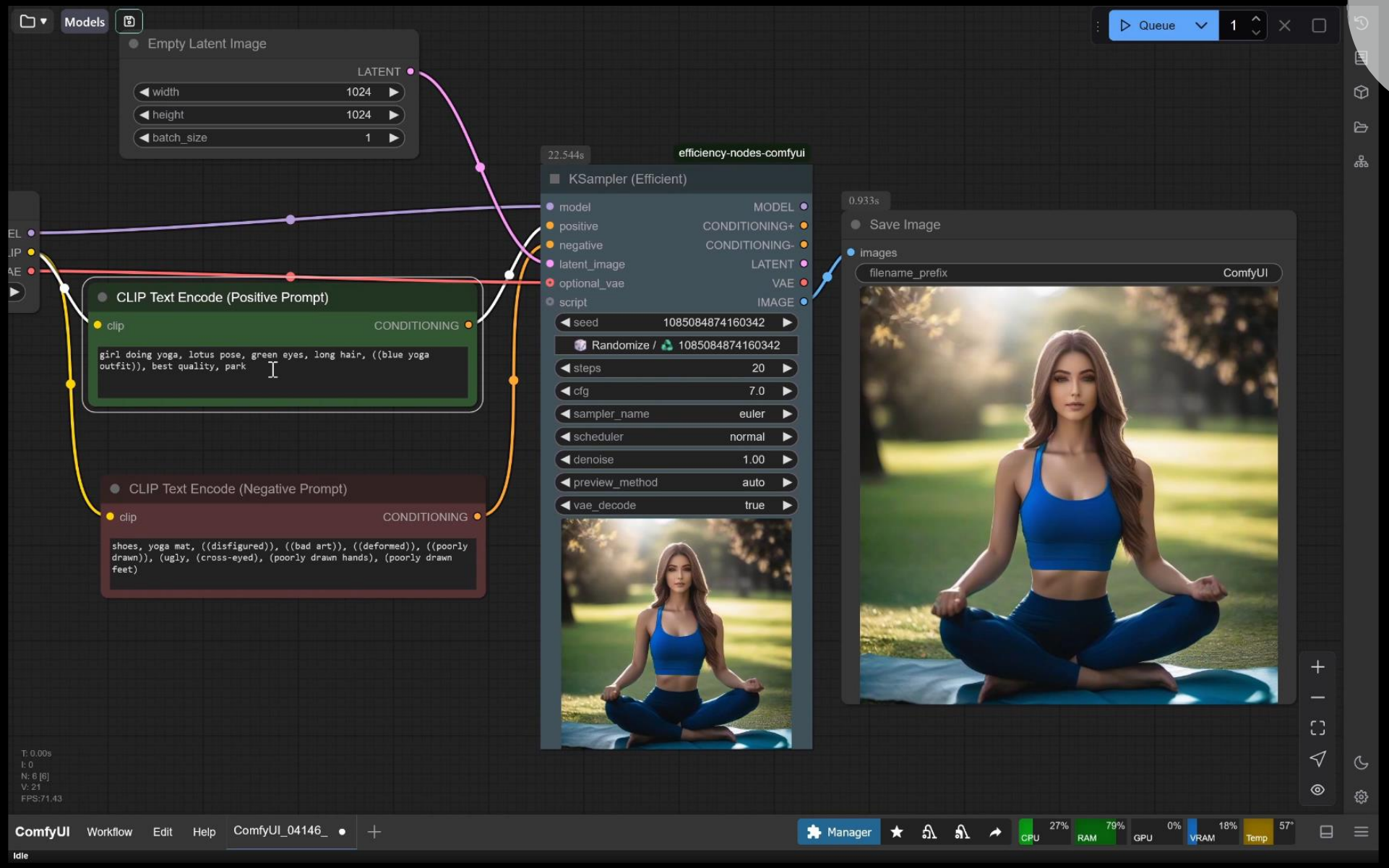
mid long dark hair,
long white t-shirt,
((blue yoga pants))

Embeddings (Textual Inversions)

- **Vector Representations of Text**
  - "Instructions" for image generation
  - Style, theme, texture, pose, character characteristics, etc.
- **Files containing additional concepts**
  - Injected into the prompt
  - The community provides many presets
  - Must be aligned with the Model version



Demo – T2I + Embedding (Textual Inversion)



Embeddings (Textual Inversions)





3D Render



Comic



Fantasy



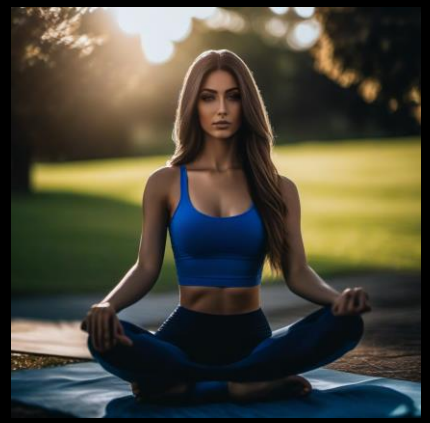
Ghibli



Vector Art



Analog Film



Cinematic



Cyberpunk



Digital Art

We can tweak the style
But no control on the pose...



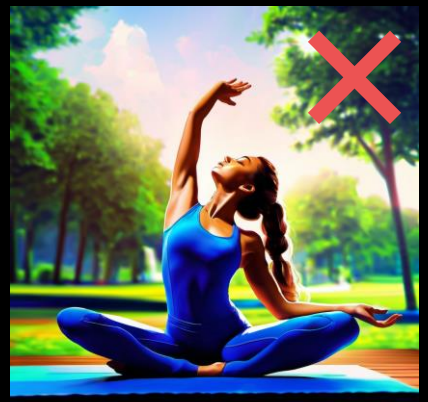
Lotus
pose



Tree
pose



Camel
pose



Shoulder-stand
pose

Text-to-Image generation is not enough!

Let's move on to Image-to-Image generation

Generate an image of a woman doing yoga **based on an existing image**

Demo – Image-to-Image (I2I)

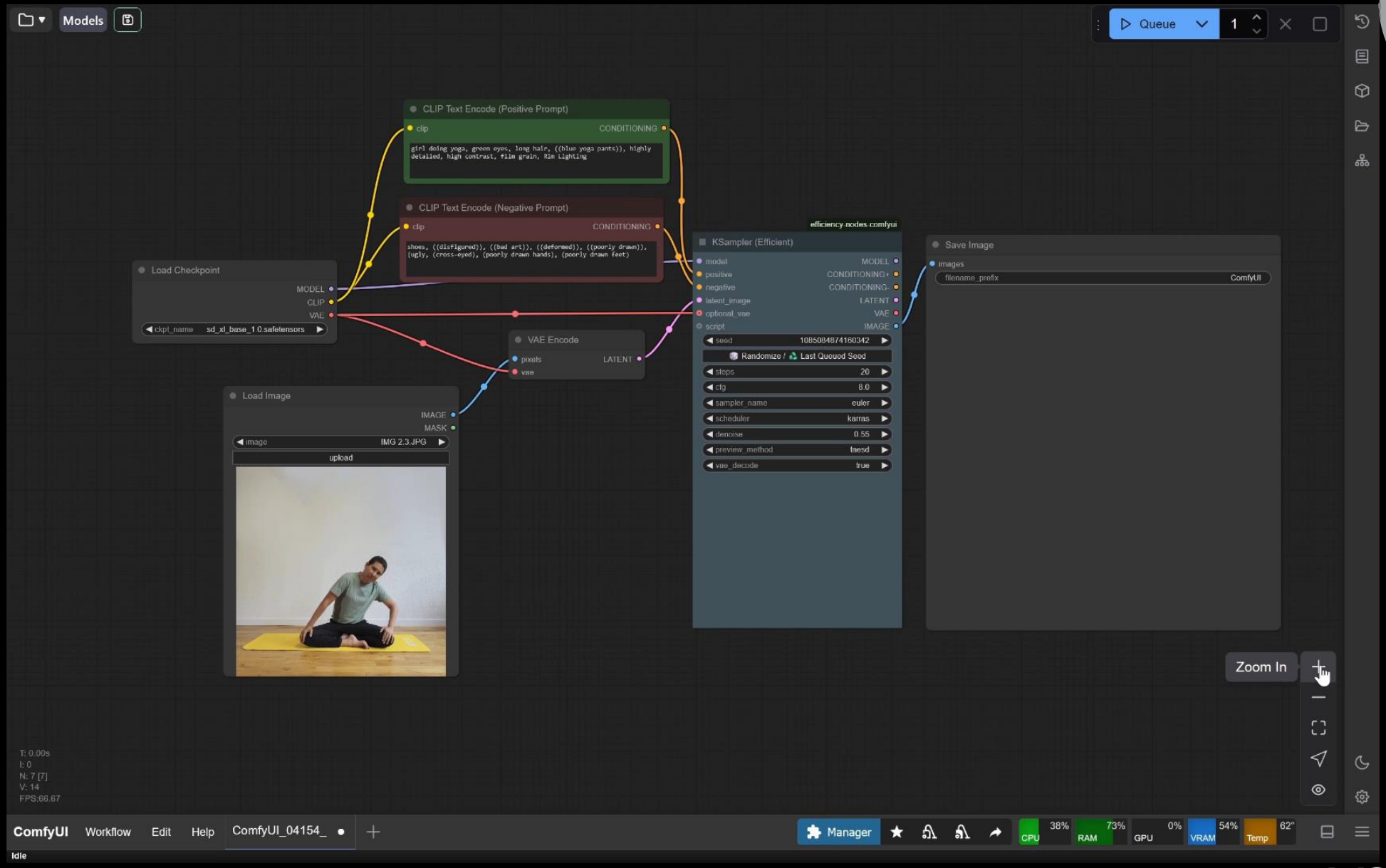
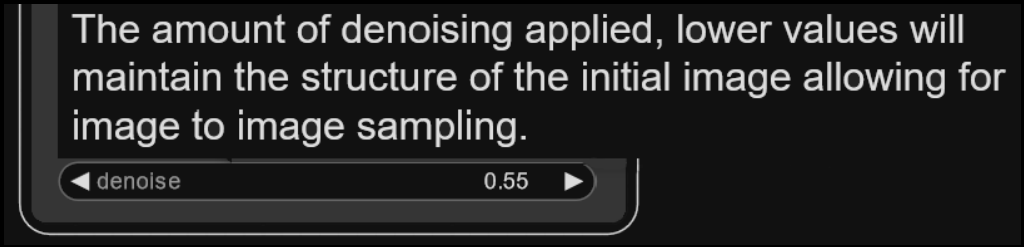
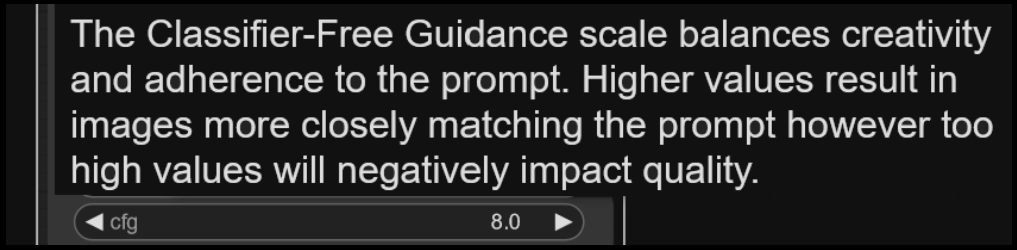


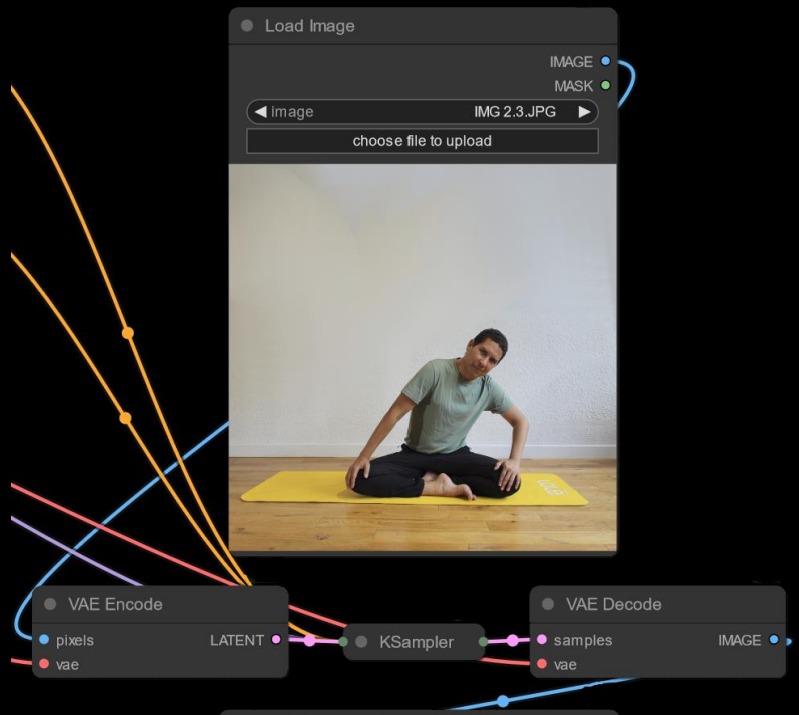
Image-to-Image Generation

- **Input Image**
  - Real image rather than latent image
  - VAE encoding (from the model)
  - We play with the % of denoising



  - The prompt has less impact
- **Increasing CFG reduces quality**





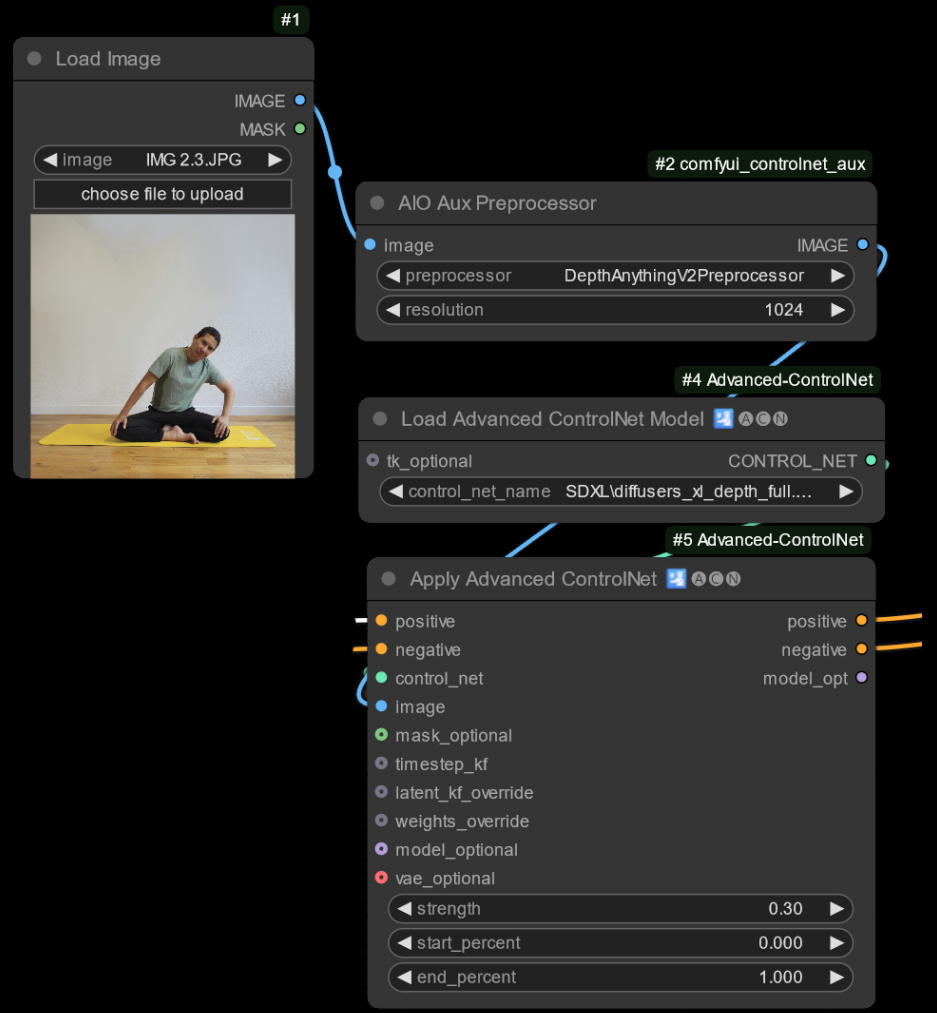




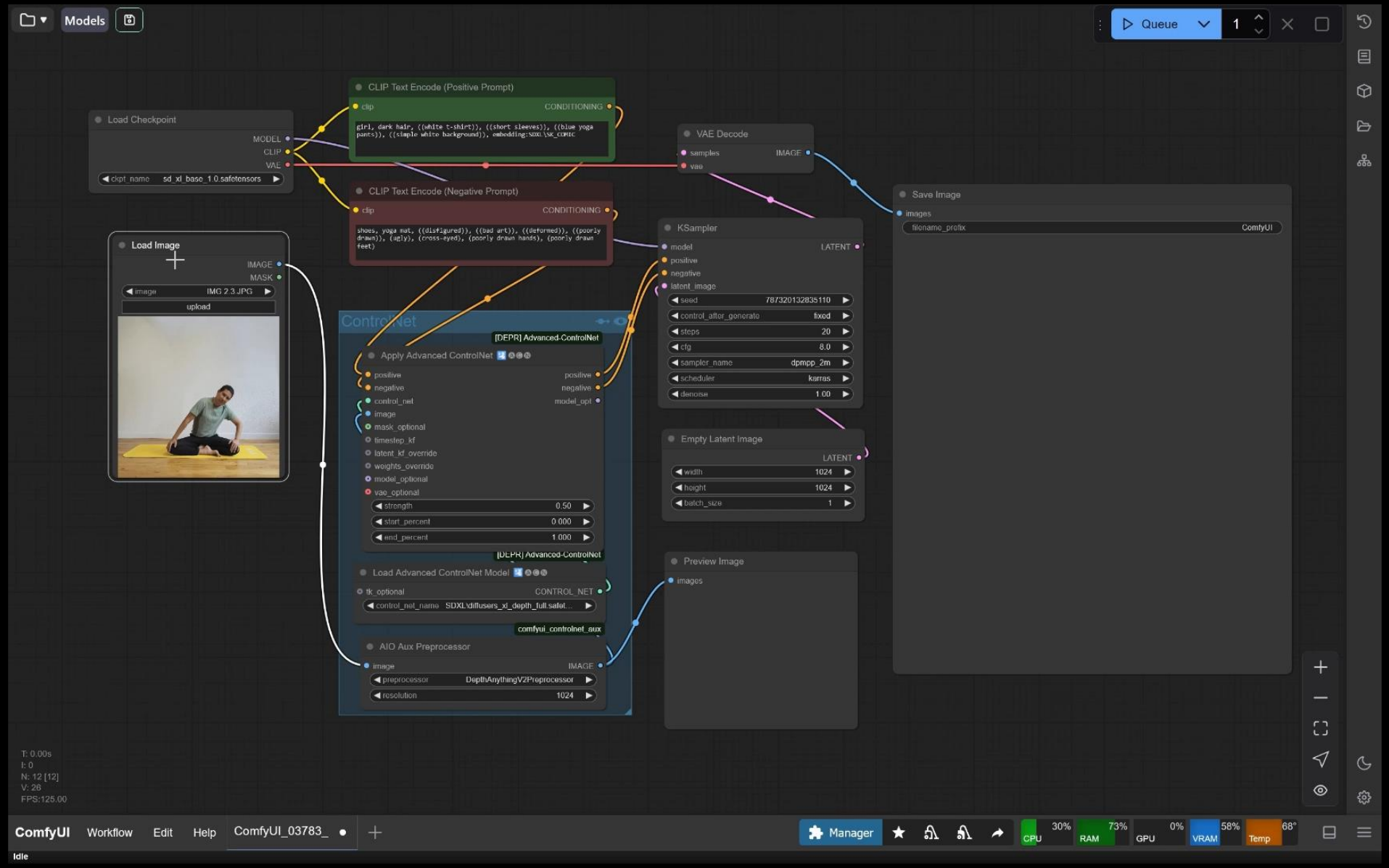


ControlNets

- **Specialized Neural Networks**
  - Additional control and guidance
  - Transfer of reference image features
  - → Hybrid approach with textual/visual references
- **Control methods**
  - Structure: pose, edges, segmentation, depth
  - Texture and detail: sketching, stylizing from edges
  - Content and layout: bounding boxes, masks
  - Abstract and style: color maps, texture fields



Demo – I2I + ControlNet



Examples of ControlNet Preprocessors



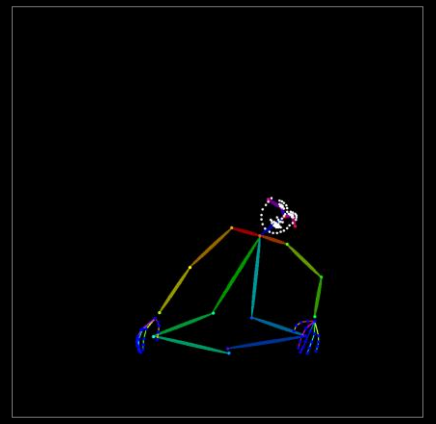
Initial Image



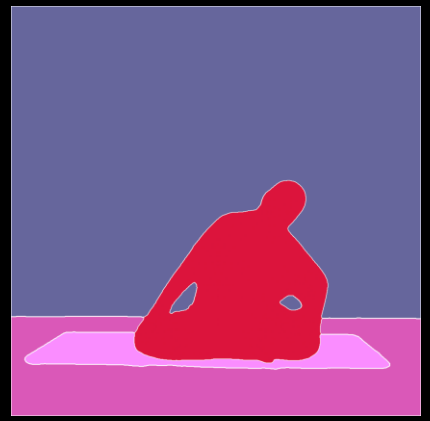
Line Art



Color Map



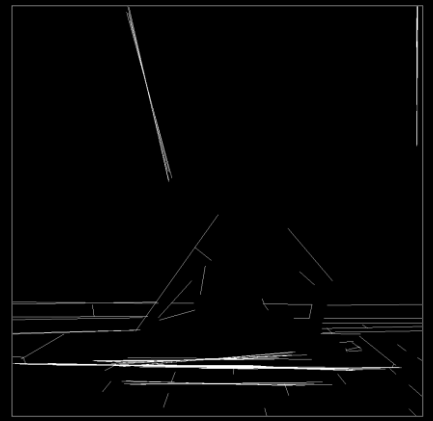
Open Pose



Segmentation



Depth Map



Straight Lines

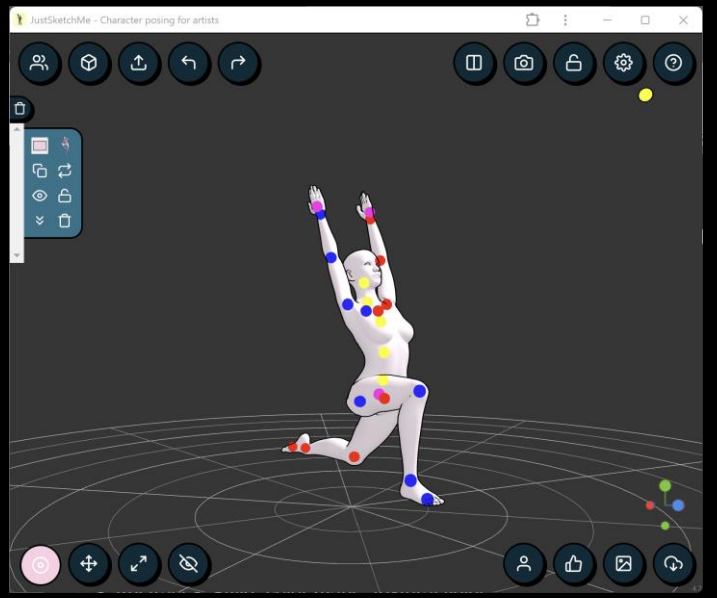


Scribble

More abstract input images

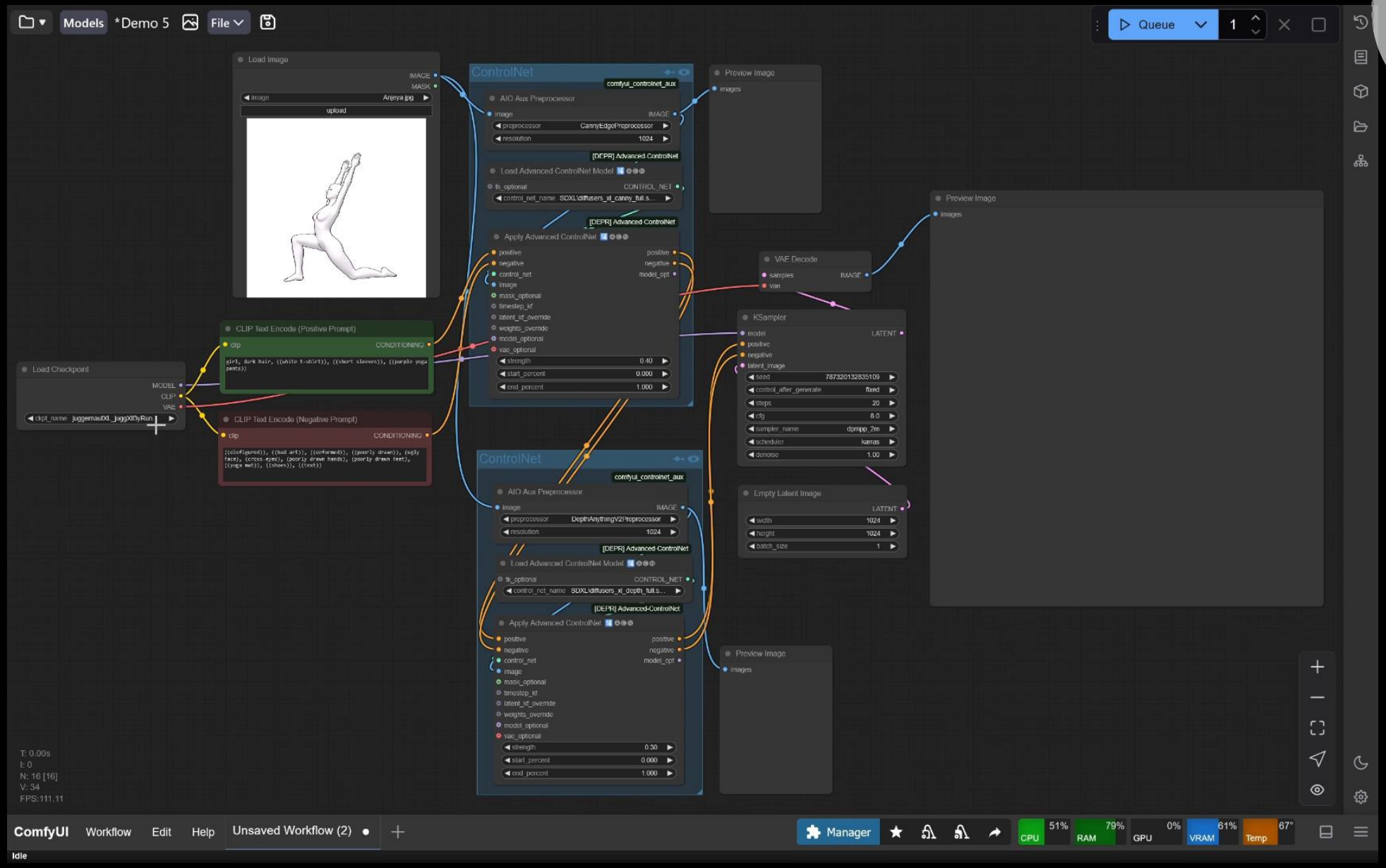
- **3D poses exported as an images**
  - Using the JustSketchMe tool (webapp & PWA)
  - Modeling based on my own knowledge
  - Multiple viewing angles





JustSketchMe

Demo – T2I + 2 ControlNets



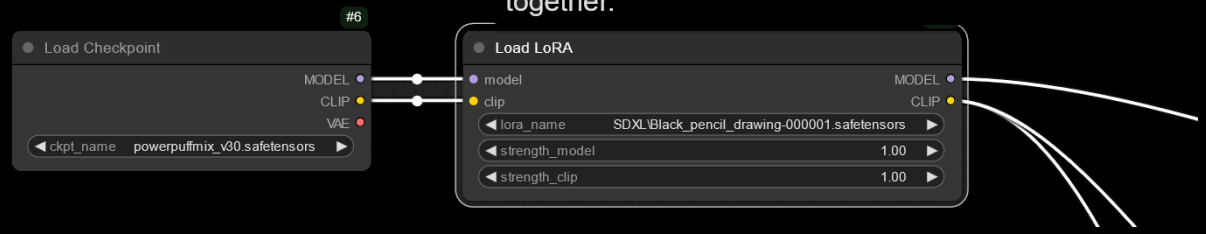
How to get more consistency within a sequence?

Generate a series of images with the same **identity**

LoRA

- **Low-Rank Adaptation**
  - Slight adaptation of the basic model
  - Changes a small subset of the parameters
- **Highly effective**
  - Smaller files, use less memory
  - Faster training
- **Usage**
  - Styles, poses, characters, or concepts
  - Triggered by keywords in the prompt
  - Numerous LoRAs provided by the community

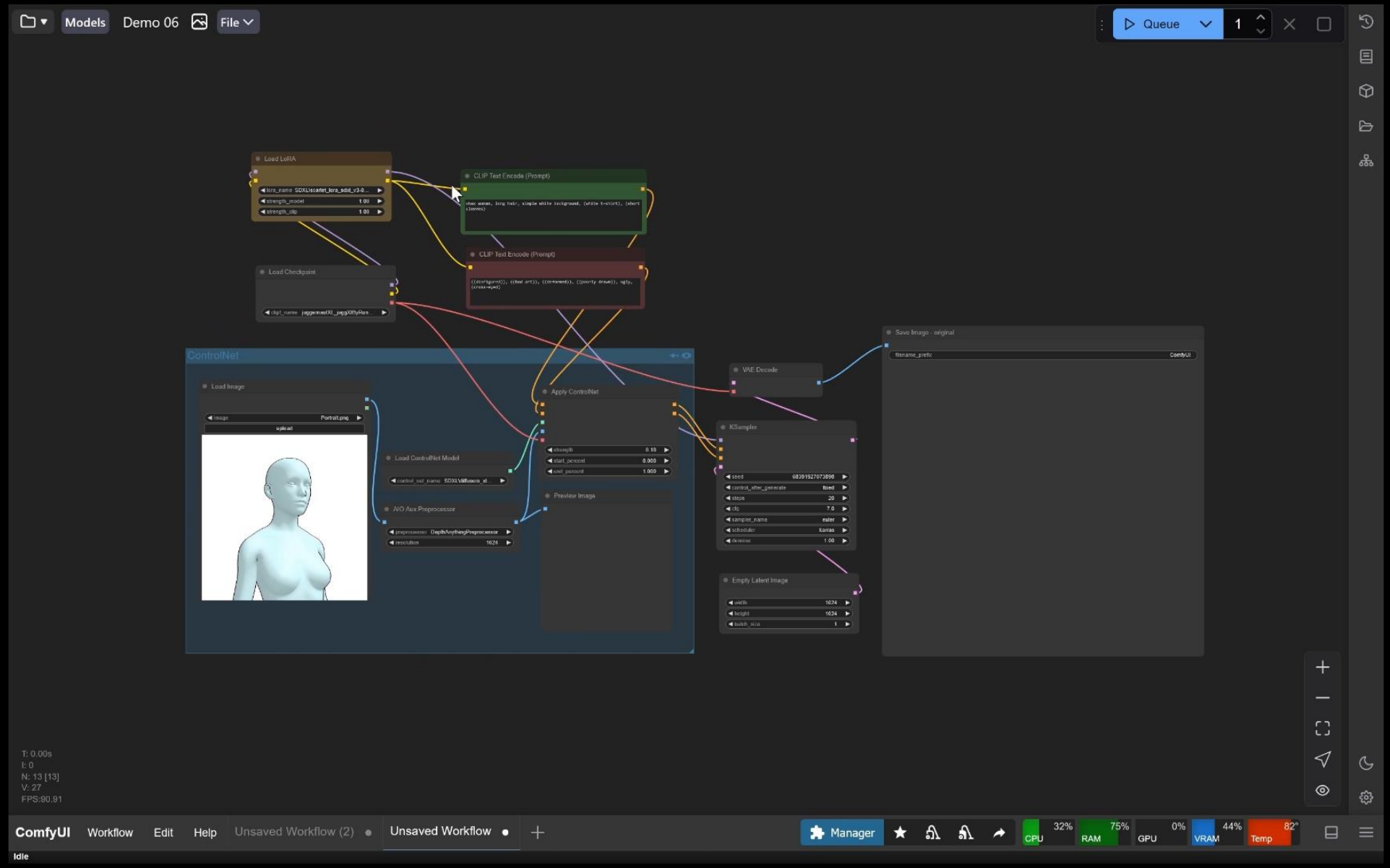
LoRAs are used to modify diffusion and CLIP models, altering the way in which latents are denoised such as applying styles. Multiple LoRA nodes can be linked together.



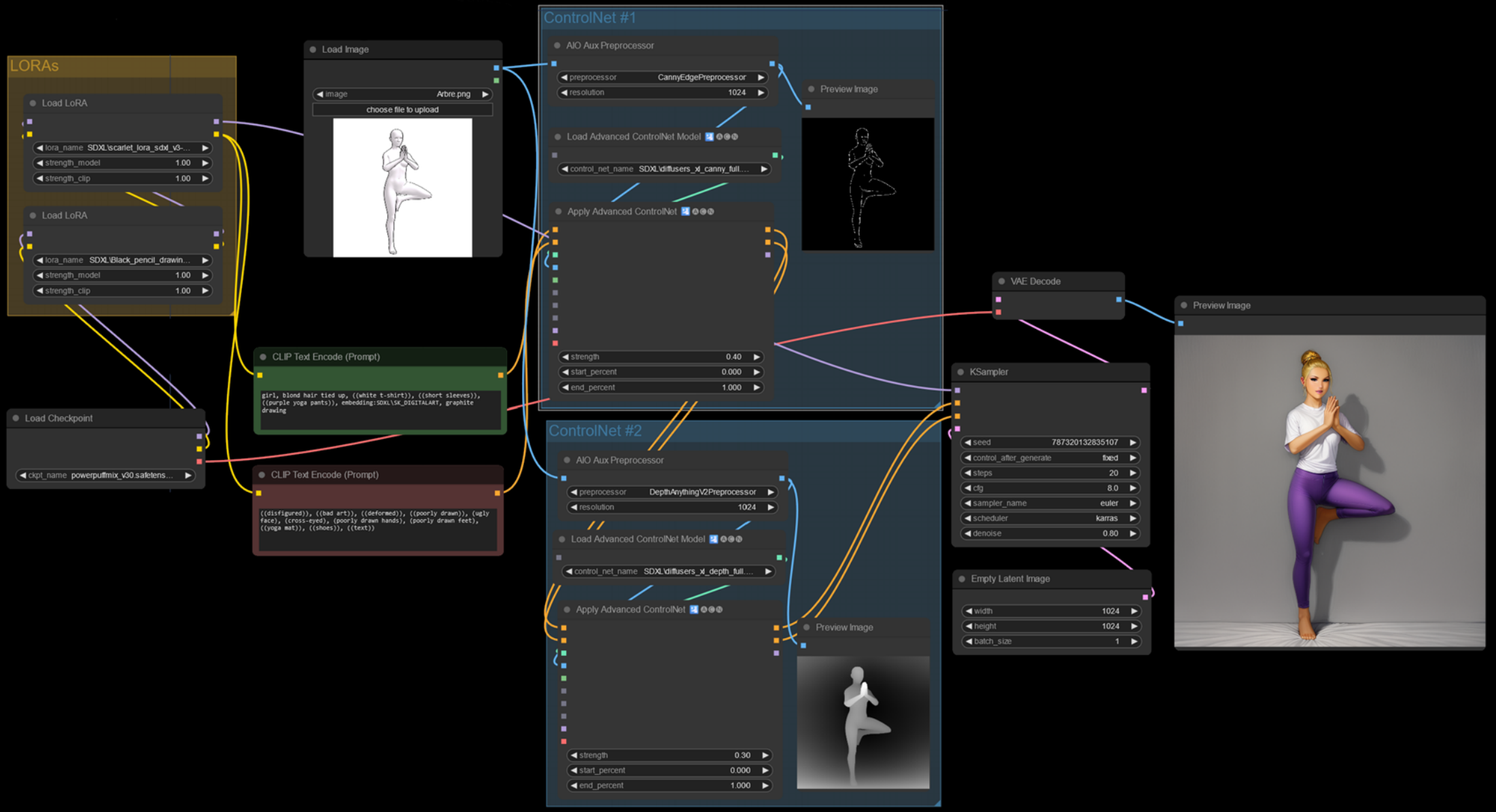




Demo – ControlNet + LoRA



Embeddings + 2 ControlNets + LoRA



Embeddings + 2 ControlNets + LoRA



Embeddings + 2 ControlNets + LoRA



Let's create our own Haj3r LoRA

Dreambooth

- Based on SDXL
- 28 input images
- 2 epochs
- Google Colab Notebook
- 1:30 (remote)





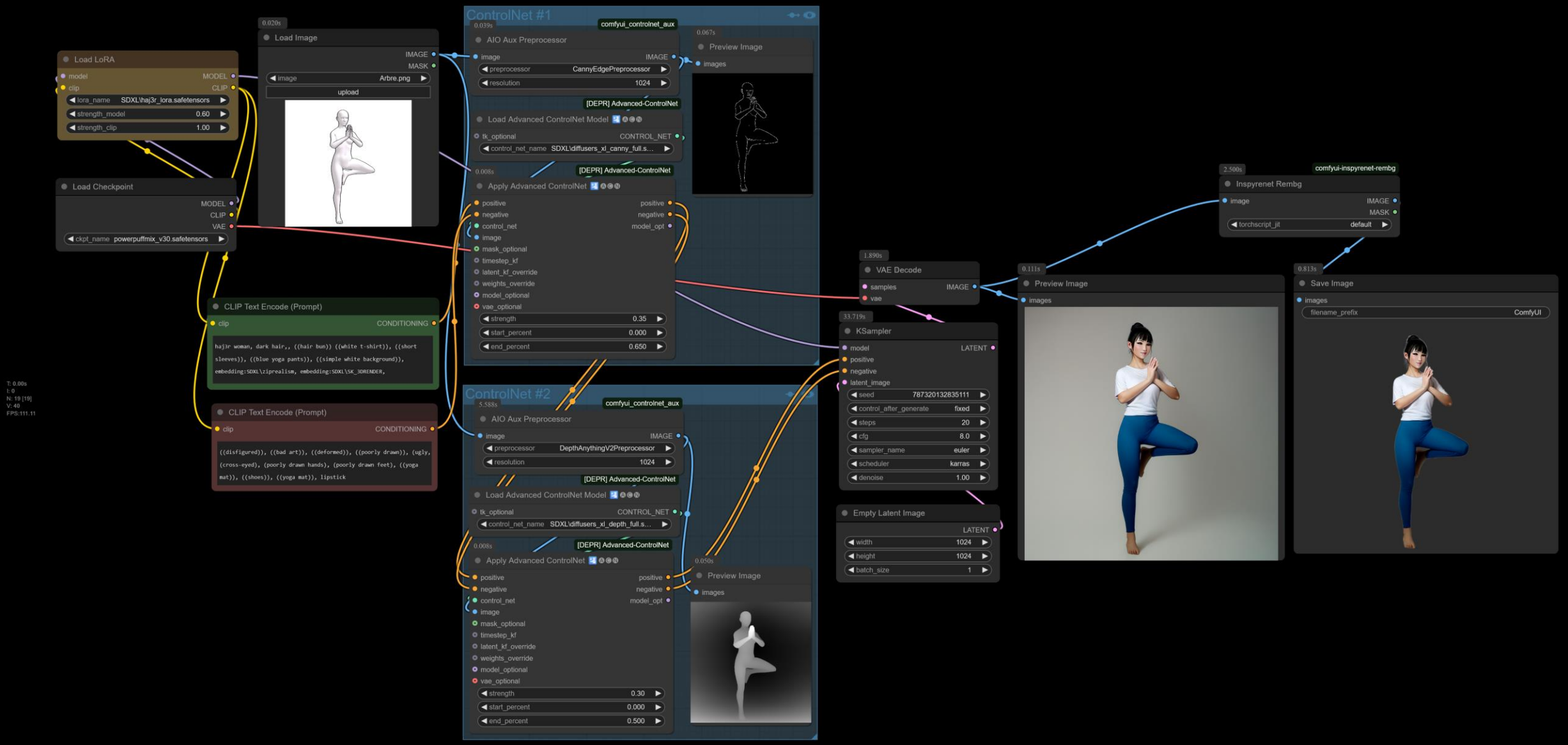
Kohya_ss

- Based on PowerPuffMix
- 15 input images
- 20 epochs
- 3:30 (local)
- Embeddings





t2i + Embeddings + ControlNets + LoRA Haj3r

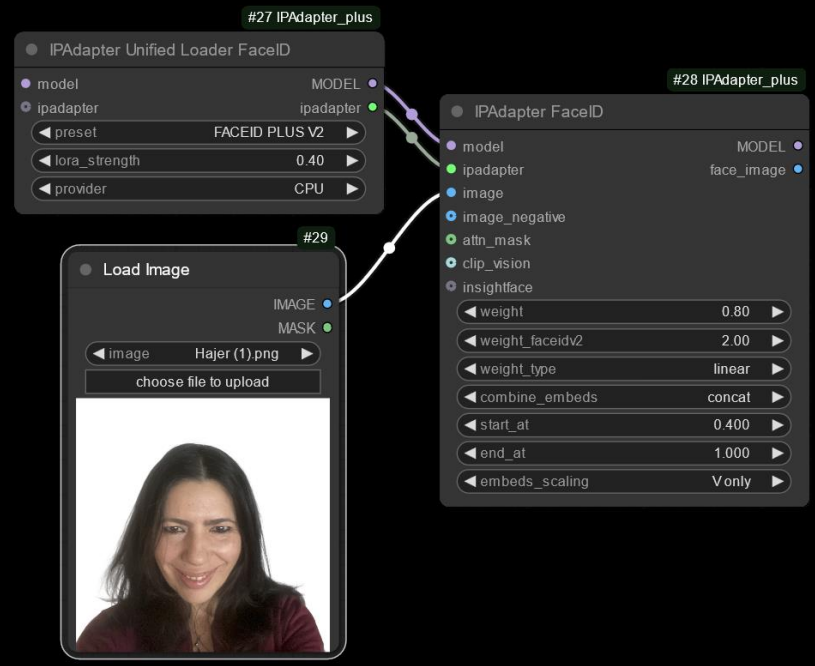


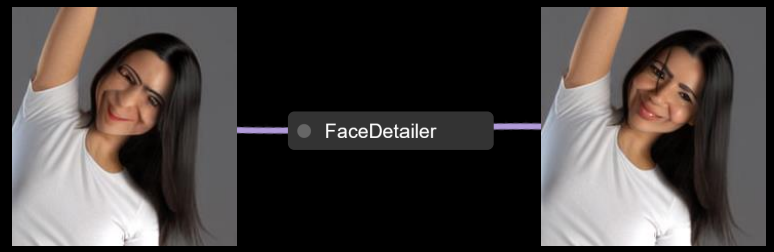
t2i + Embeddings + ControlNets + LoRA Haj3r



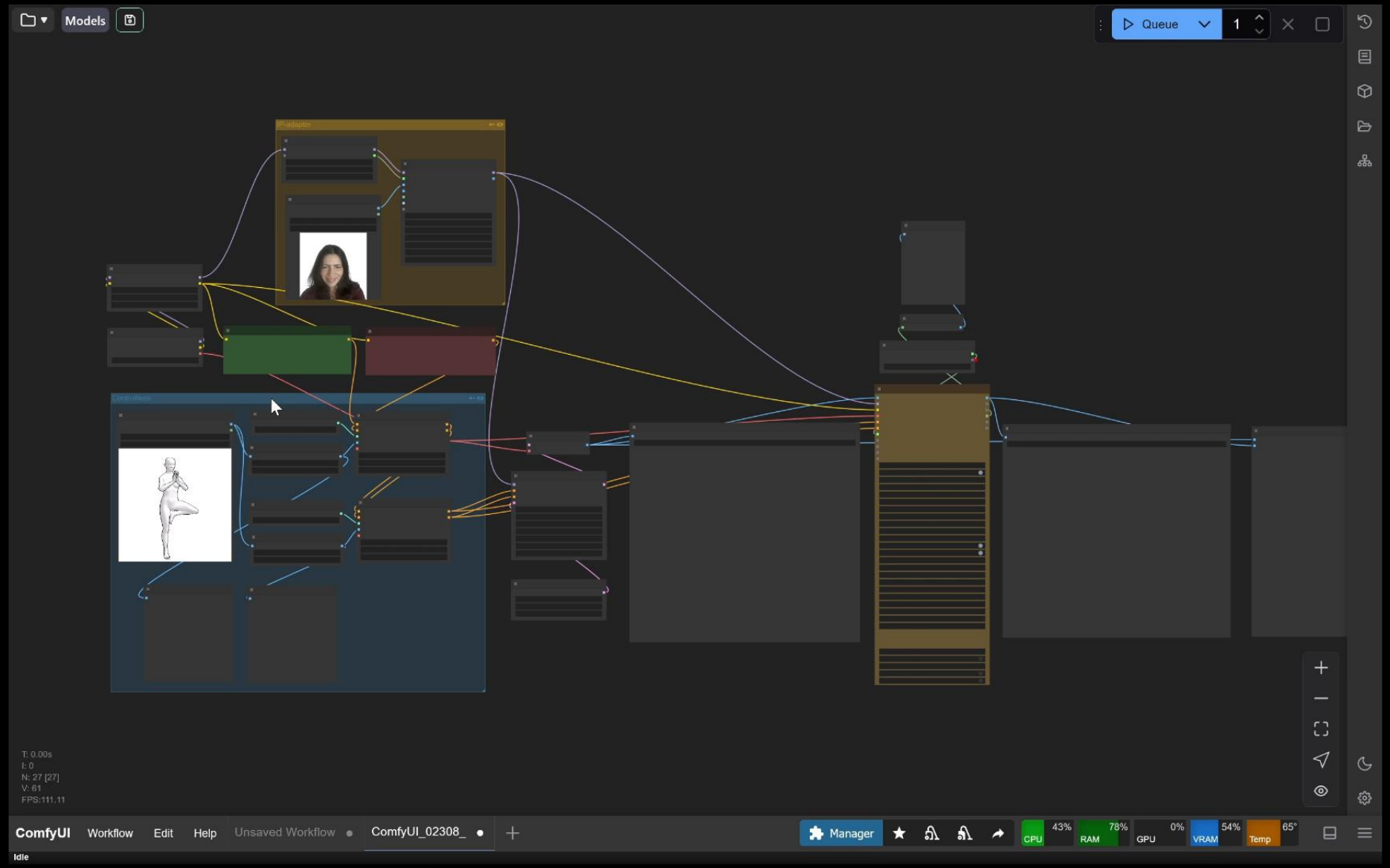
FaceID + FaceDetailer

- **Image Prompt Adapters**
  - Generation from visual prompts
  - Pre-trained control networks, different from SD
  - A kind of LoRA with an image
- **FaceID IPAdapter**
  - Facial recognition model
  - LoRA to improve identity consistency
- **FaceDetailer**
  - Facial enhancement (eyes, nose, lips, expression)
  - Post-processing AI model

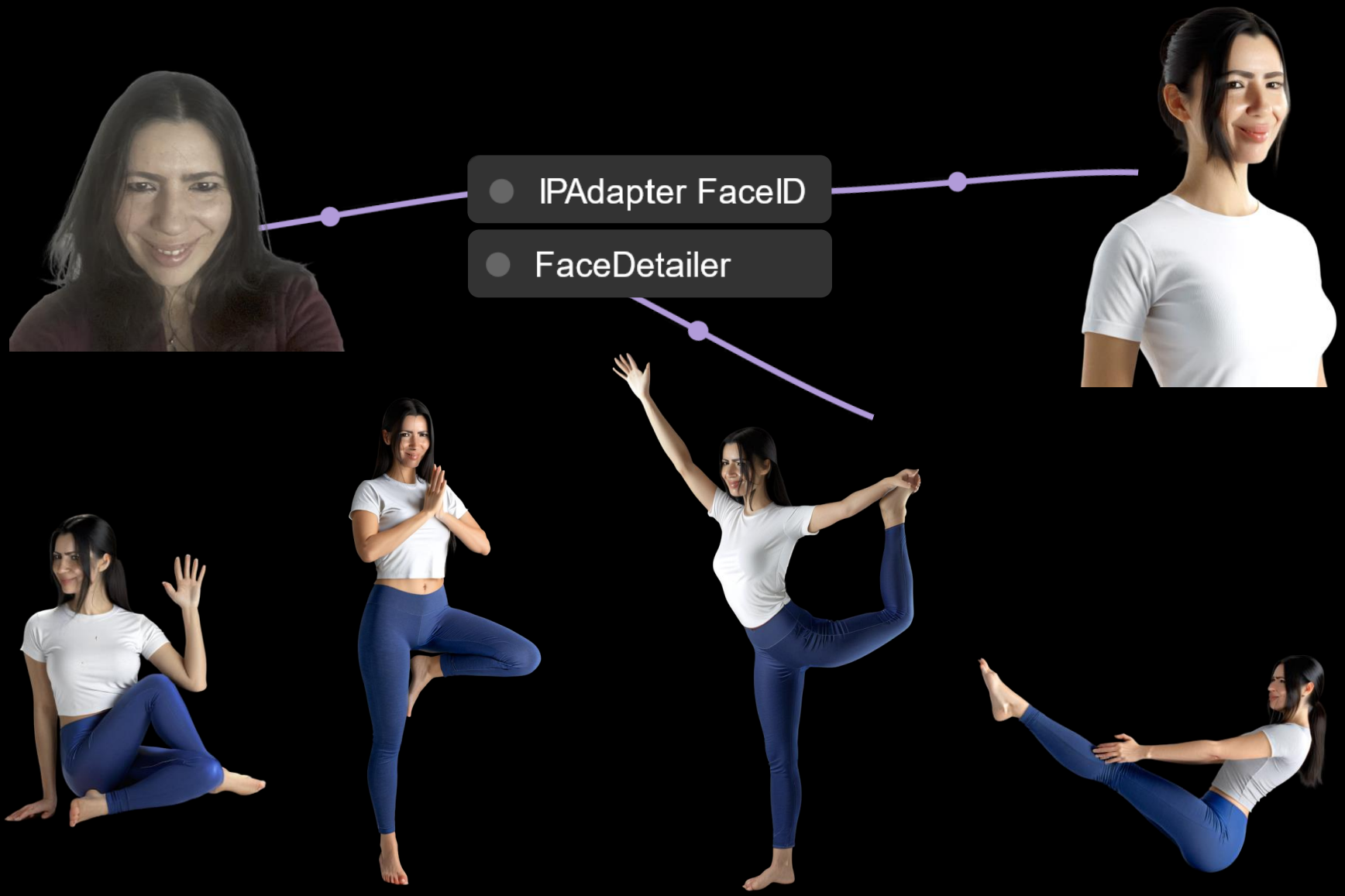




Demo – t2i + ControlNets + FaceID + FaceDetailer



Embeddings + ControlNets + FaceID + FaceDetailer



Easier to switch models

Cheyenne v2



Easier to switch personas



Let’s summarize



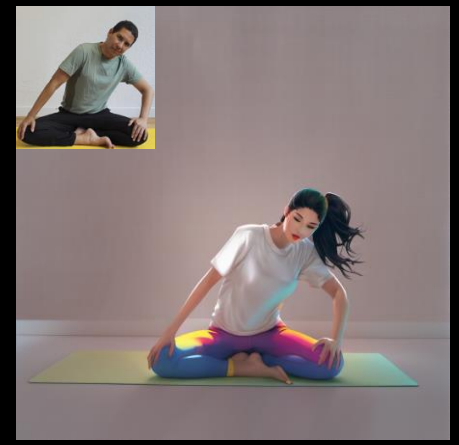
t2i



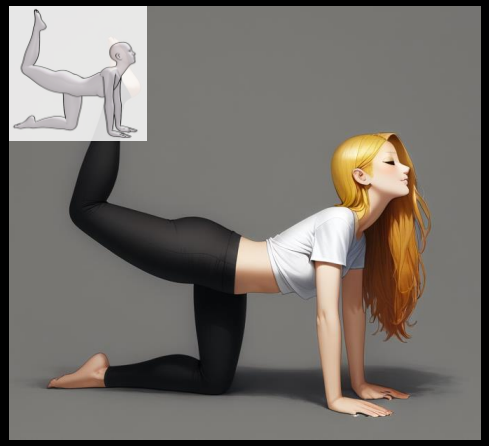
t2i + embeddings



t2i + i2i



t2i + embeddings + ControlNet



t2i + embeddings + ControlNets + LoRA



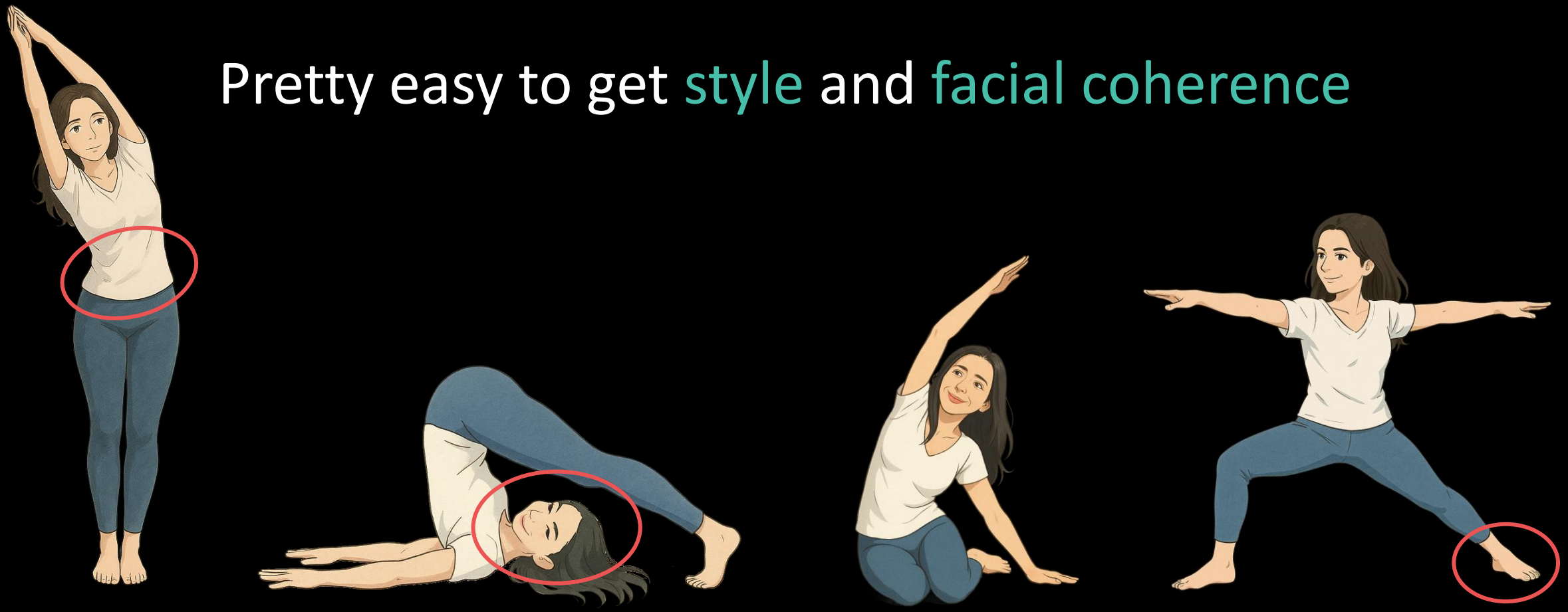
t2i + embeddings + ControlNets + LoRA Haj3r



t2i + embeddings + ControlNets + FaceID + FaceDetailer

Wait! What about OpenAI GPT-Image-1?

Pretty easy to get style and facial coherence



But no fine control on the poses

And Google Gemini / Imagen?

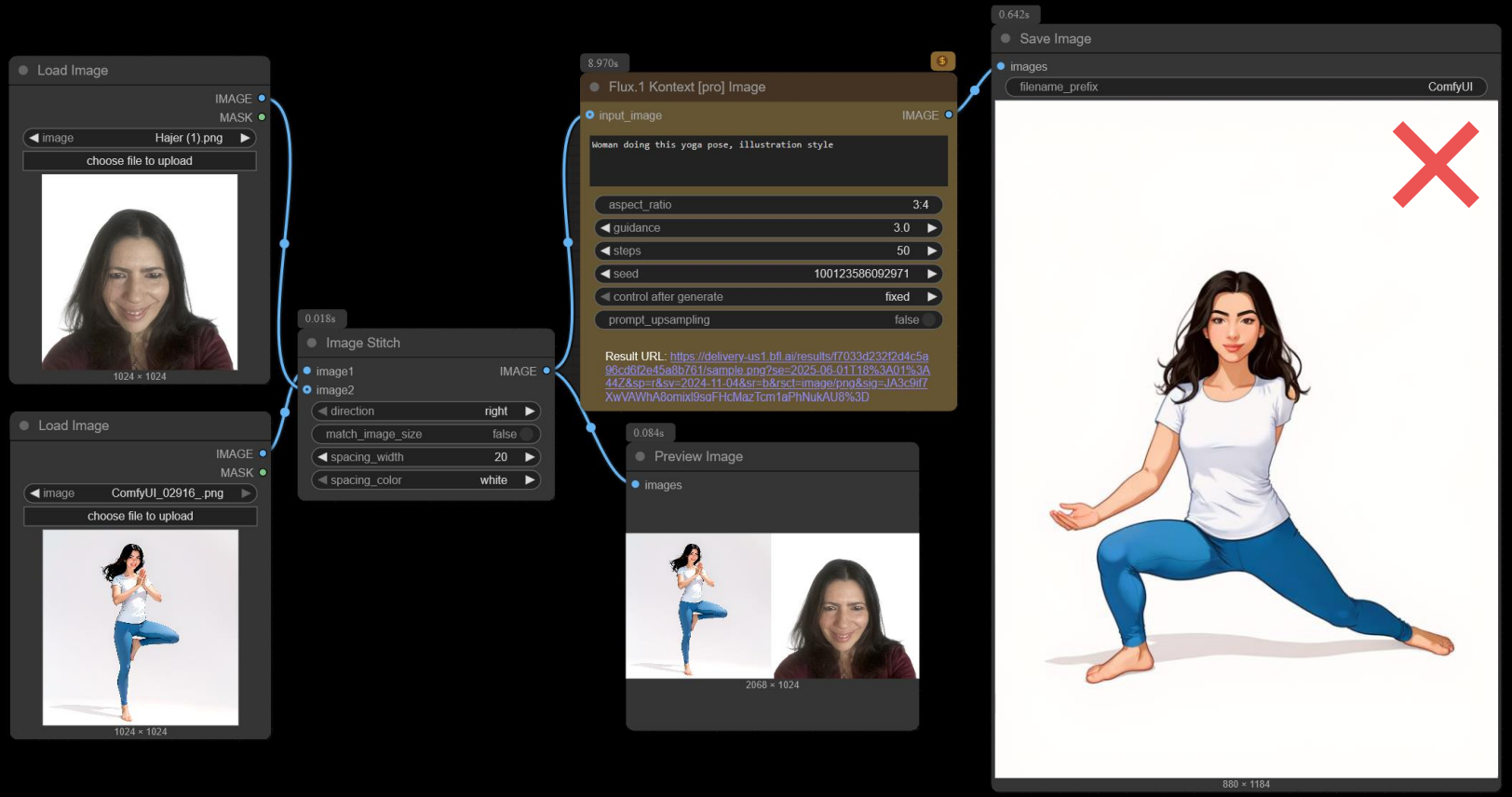


Draw an image of a beautiful woman with long dark hair, wearing a white t-shirt with v-neck and short sleeves and blue yoga pants, doing the exact same yoga pose as the input image, simple white background



No control at all on the poses

Or the brand-new Flux Kontext?



No control on both poses and face

My initial use case is implementable

Accurate and consistent images

Image generation is a both Science & Art

Numerous components / parameters

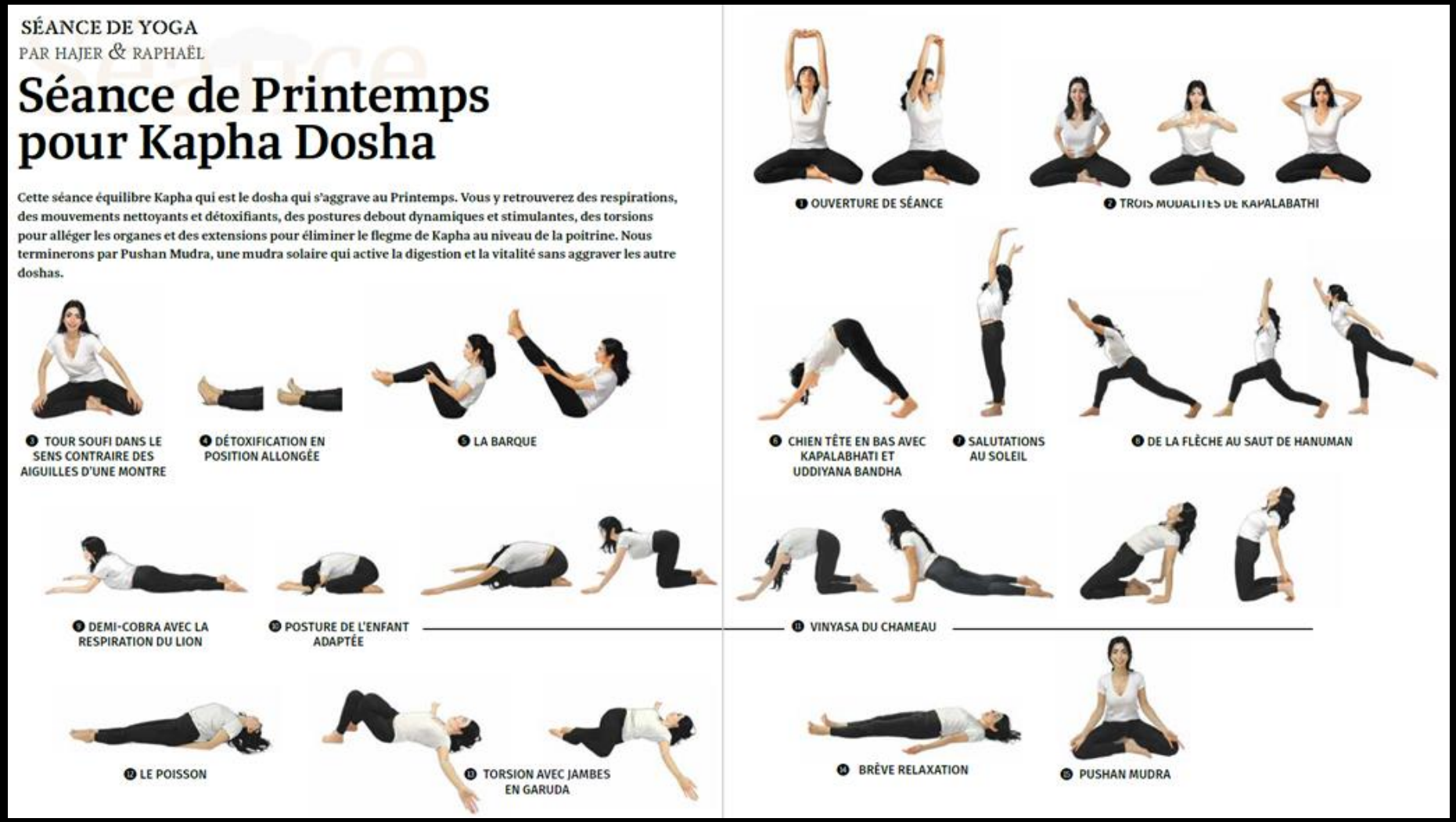
Icing on the cake: more **inclusivity**



First concrete result

LE JOURNAL DU yoga

#261
March-April 2025



SÉANCE DE YOGA

PAR HAJER & RAPHAËL

Séance de Printemps pour Kapha Dosha

Cette séance équilibre Kapha qui est le dosha qui s'aggrave au Printemps. Vous y retrouverez des respirations, des mouvements nettoyants et détoxifiants, des postures debout dynamiques et stimulantes, des torsions pour alléger les organes et des extensions pour éliminer le flegme de Kapha au niveau de la poitrine. Nous terminerons par Pushan Mudra, une mudra solaire qui active la digestion et la vitalité sans aggraver les autre doshas.

Next step… video?

Comfy

Wan



The Yoga of Image Generation



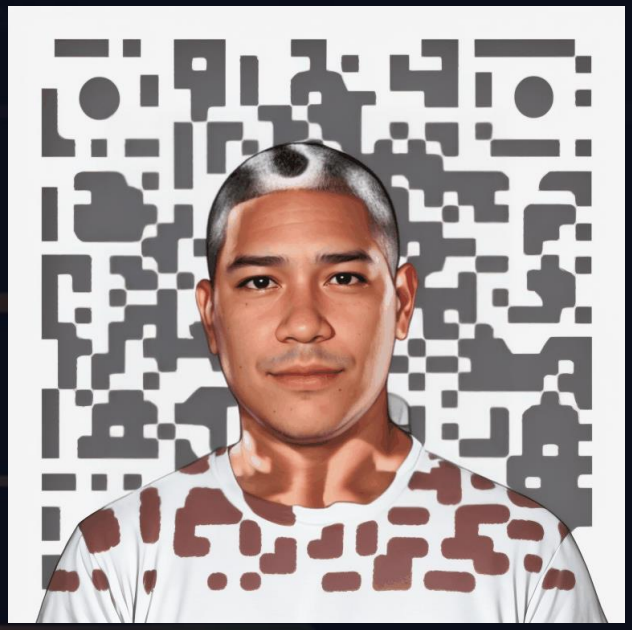
स्थिरसुखमासनम्

Yoga Sūtra II.46

The Pose should be Stable and Comfortable

Thank you!

Your Feedback



raphiki.github.io

DEVOXX™ POLAND